

System Flashing

System Flashing

1. Preparations
2. Download Image
3. Flash System
4. Boot System
5. Set Login Mode

1. Preparations

Before flashing the Ubuntu system image, the following preparations are required:

- Prepare a Micro SD card with at least 32GB capacity
- SD card reader
- Formatting tool: SDFormatter
- Tool for flashing images on Windows system: Win32DiskImager

You can download the flashing system tools from the download section in the documentation link

RDk X5 MagicBox

- 1.RDK MagicBox ▾
- 2.RDK MagicBox Introduction ▾
- 3.RDK MagicBox User Guide ▾
- 4.RDK MagicBox Basic Settings ▾
- 5.RDKX5 Magicbox Basic Development Course ▾
- 6.Online AI Model Course ▾
- 7.OpenClaw Embodied AI Development ▾
- 8.ROS2 Basic Tutorial ▾

Download

Code

Software_tools

SystemFile

Welcome to RDk X5 MagicBox repository

The OpenClaw + AI large model

RDk X5 MagicBox



10 TOPS
8GB RAM

✓ Built-in Rich Vision Algorithms ✓ Integrated Large Model Voice Interaction

2. Download Image

The RDk kit currently provides Ubuntu 22.04 system image, which supports Desktop graphical interaction.

Before using the RDk X5 development board, you need to flash the system first. The system image provided in our online repository is modified based on the official RDk MagicBox image version from RDk.

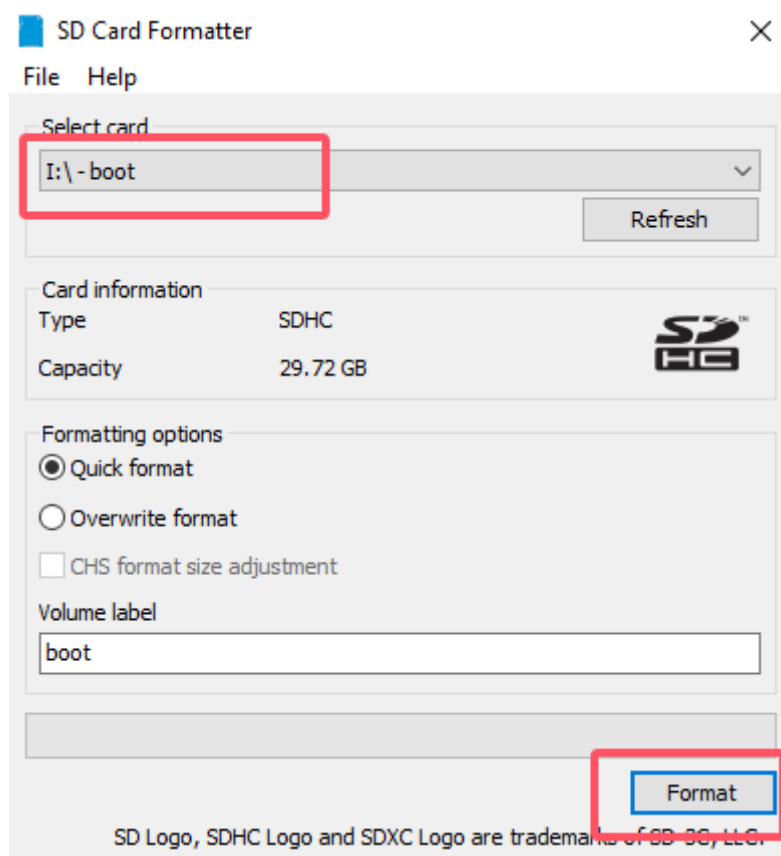
Go to the image directory in the online repository to download our Ubuntu system with desktop, the suffix is .img compressed package, and extract it for later use.

If you want to download the RDK MagicBox clean image, please refer to the link below and select the first option

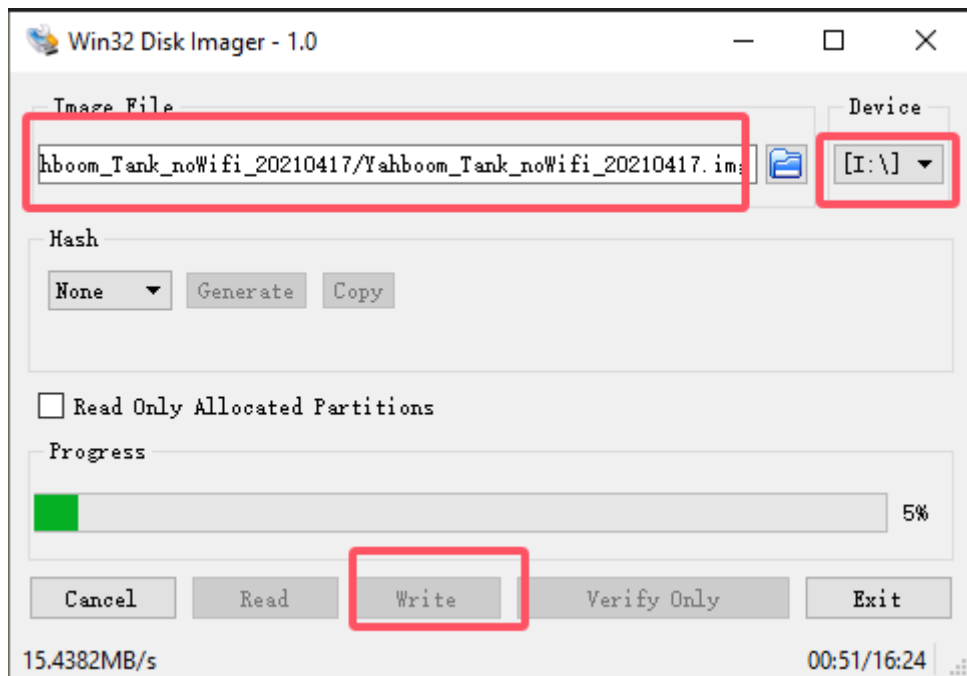
Download Link	Description
RDK OS	RDK OS Image

3. Flash System

1. Insert the SD card into the card adapter or card reader, then connect to the computer;
2. Use the dedicated formatting tool SDFormatter to format the memory card



3. Extract and run the win32diskimager tool. In the software, select the img (image) file, select the SD card drive letter under "Device", then select "Write" to begin flashing the system.



4. After flashing is complete, a dialog will pop up indicating completion. If unsuccessful, please close firewall and similar software, reinsert the SD card and try again. Note: after flashing, you will see the SD card only shows 74MB in Windows. This is normal because the Linux disk partitions are not visible under Windows!

Note: After successful flashing, the system may prompt to format the memory card. Please do not format the flashed memory card.

4. Boot System

First, keep the development board powered off. Then insert the prepared storage card into the Micro SD card slot on the development board. Connect the development board to the monitor via HDMI cable, and finally power on the development board.

Note: Development Board Indicator Light Description

white lights turn on sequentially, wait approximately 50s

white light flashing with sound effect: Startup complete

After the Ubuntu Desktop version system finishes booting, it will output the system desktop on the monitor through the HDMI interface, as shown in the figure below:

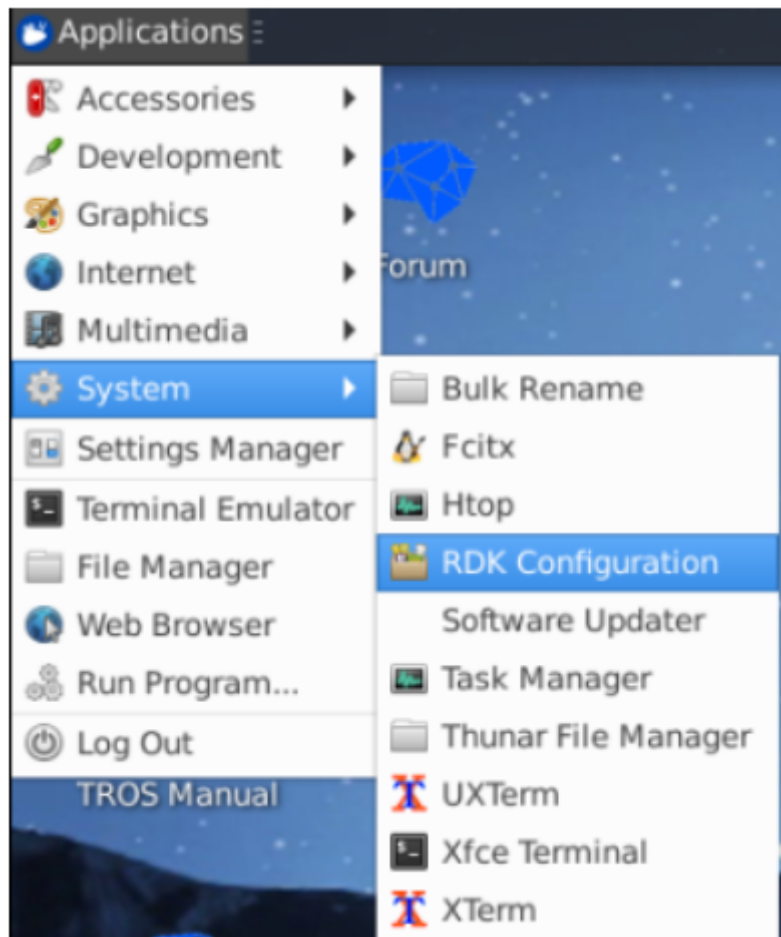


5. Set Login Mode

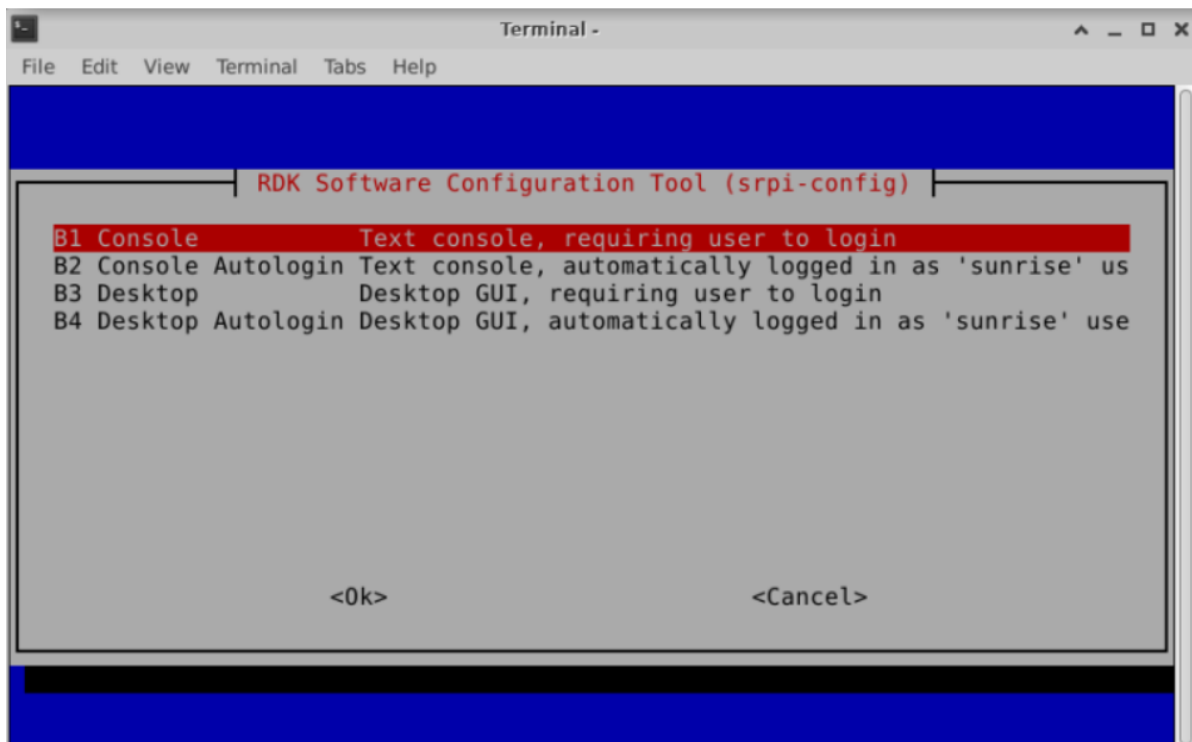
The desktop graphical system supports four login modes:

1. Enable graphical interface, auto login
2. Enable graphical interface, manual user login
3. Character terminal, auto login
4. Character terminal, manual user login

Find the RDK Configuration item in the menu bar and click to open.



Select System Options -> Boot / Auto Login to enter the following configuration items. Select the appropriate option according to your needs.



The changes will take effect after restarting.

